NUET Mock №5

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Mathematics

1. Water makes up 80 per cent of fresh mushrooms. However, water makes up only 20 per cent of dried mushrooms. By what percentage does the mass of a fresh mushroom decrease during drying?

   A. 60
   B. 70
   C. 75
   D. 80
   E. 85

Ans: C

2. Two circles have the same radius.

   The centre of one circle is (-2, 1).

   The centre of the other circle is (3, -2).

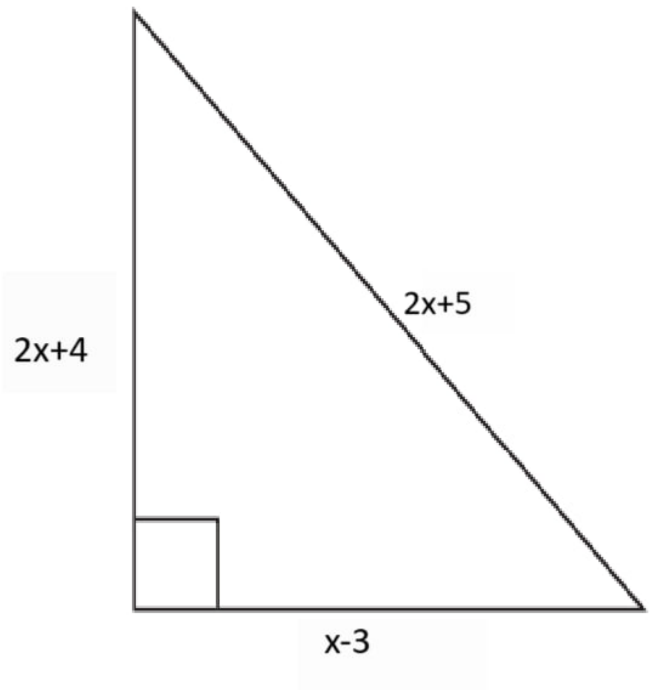
   The circle intersect at two distinct points.

   What is the equation of the straight line through the two points at which the circles intersect?

   A. $3x - 5y = 4$
   B. $3x + 5y = -1$
   C. $5x - 3y = -4$
   D. $5x - 3y = -1$
   E. $5x - 3y = 1$
   F. $5x - 3y = 4$
   G. $5x + 3y = 1$

Ans: F

3. The diagram shows a right-angled triangle, with sides of length $x - 4$, $2x + 4$ and $2x + 5$, all in cm.



What is the area, in cm^2 of the triangle?

A. 10
B. 12
C. 28
D. 36
E. 40
F. 54
G. 70
H. 84

Ans: H

4. There are three towns: Astana, Taraz, Kyzylorda.
The distance from Astana to Taraz is the same as the distance from Astana to Kyzylorda.
Taraz is south of Astana.
Taraz is on a bearing of 110° from Kyzylorda.
What is the bearing of Kyzylorda from Astana?

A. 040°

B. 070°
C. 110°
D. 140°
E. 180°
F. 210°
G. 220°
H. 240°

Ans: G

5. Which of the following is a simplification of

$$4-\frac{x(3x+1)}{x^2(3x^2-2x-1)}?$$

A. $\frac{12x^3-8x^2-7x-1}{x(3x-1)(x-1)}$
B. $\frac{4x^2+4x-1}{x(x+1)}$
C. $\frac{4x^2+4x+1}{x(x+1)}$
D. $\frac{4x^2-4x-1}{x(x-1)}$
E. $\frac{4x^2-4x+1}{x(x-1)}$
F. $\frac{12x^3-8x^2-x+1}{x(3x-1)(x-1)}$

Ans: D

6. P, Q and R are regular polygons.
Q has two times as many sides as P.
An interior angle of Q is 20° larger than an interior angle of P.
R has twice as many sides as Q.
How much larger is an interior angle of R than an interior angle of Q, in degrees?

A. 1
B. 1.5
C. 2
D. 2.5

E. 3
F. 5
G. $6\frac{2}{3}$
H. 10

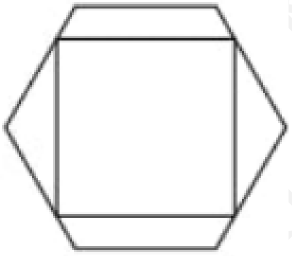
Ans: C

7. Consider the equation $2x^2 + 4x + c = 0$, where c is constant.
The positive difference between the roots of this equation is $\sqrt{10}$.
What is the value of c?

A. -5
B. -4.5
C. -3
D. -0.5
E. 0.75
F. 8

Ans: C

8. A square has its vertices on the edges of a regular hexagon. Two of the edges of the square are parallel to two edges of the hexagon as shown in the diagram. The sides of the hexagon have length 1

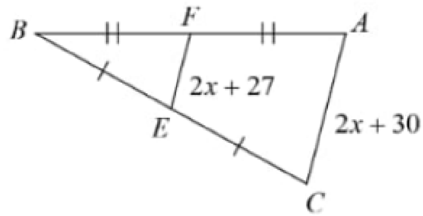


What is the length of the sides of the square?

A. $\frac{5}{4}$
B. $3 - \sqrt{3}$
C. $\frac{4}{3}$
D. $\sqrt{2}$
E. $\frac{3}{2}$

Ans: B

9. Find the value of x.



   A. -12
   B. -6
   C. 5
   D. 10
   E. 8

Ans: A

10. The positive integers m, n and p satisfy the equation $3m + \frac{3}{n+\frac{1}{p}} = 17$

    What is the value of p?

   A. 9
   B. 6
   C. 4
   D. 3
   E. 2

Ans: E

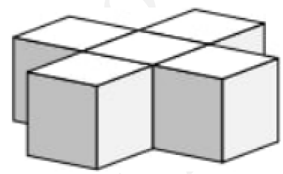
11. Used in measuring the width of a wire, one mil is equal to one thousandth of an inch. An inch is about 2.5 cm.

    Which of these is approximately equal to one mil?

    A. $\frac{1}{40}$mm
    B. $\frac{1}{25}$mm
    C. $\frac{1}{4}$mm
    D. 25mm
    E. 40mm

Ans: A

12. A shape is made from five unit cubes, as shown. What is the surface area of the shape?



    A. 20
    B. 22
    C. 24
    D. 26
    E. 28

Ans: B

13. The numbers x, y and z satisfy the equation $9x + 3y - 5z = 4$ and $5x + 2y - 2z = 13$

    What is the value of $\frac{x+y+z}{3}$?

    A. 7
    B. 8
    C. 9
    D. 10
    E. 11

Ans: D

14. The lines $y = x$ and $y = mx - 4$ intersect at the point P

    What is the sum of the positive integer values of m for which the coordinates of P are also positive integers

    A. 3
    B. 5
    C. 7
    D. 8
    E. 10

Ans: E

15. The teenagers Sam and Jo notice the following facts about their ages:

    The difference between the squares of their ages is four times the sum of their ages.

    The sum of their ages is eight times the difference between their ages.

    What is the age of the older of the two?

    A. 16
    B. 17
    C. 18
    D. 19
    E. 20

Ans: C

16. Here is a pattern of numbers:

1						
2	3	4				
5	6	7	8	9		
10	11	12	13	14	15	16

The pattern of numbers is continued in the same way.
What number will appear directly below 196?

A. 218
B. 219
C. 220
D. 221
E. 222
F. 223
G. 224
H. 225

Ans: F

17. Alex, Cameron and Sam are all taking part in a 400 m race.
They are each running at a different constant speed.
Alex is running 12% faster than Cameron, whilst Sam is running 2% slower than Cameron.
When Alex crosses the finish line, how many metres is Sam from the finish line?

A. 9.6
B. 14
C. 24
D. 25
E. 28
F. 50
G. 56

Ans: F

18. The ball for a garden game is a solid sphere of volume 192 cm^3.
For the children's version of the game the ball is a solid sphere made of the same material, but the radius is reduced by 25%.
What is the volume, in cm^3, of the children's ball?

A. 48
B. 81
C. 96
D. 108

E. 144

Ans: B

19. Convert $\frac{5}{3}$ m/s to km/h.
    A. 6 km/h.
    B. 12 km/h.
    C. 15 km/h.
    D. 3.6 km/h.
    E. 5.5 km/h.

Ans: A

20. Simplify

$$3m^2n \cdot 9mn^3 \cdot (9mn^2)^{-2}$$

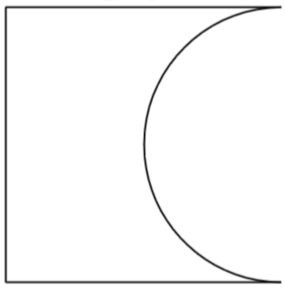
A. $3m$
B. $\frac{m}{3}$
C. $\frac{m}{9}$
D. $\frac{1}{3m}$
E. $\frac{m}{3n}$
F. $\frac{m^3}{3}$
G. $\frac{1}{3n^4}$

Ans: B

21. The perimeter of a square is $(36 - 4\sqrt{3})$ cm.
What is the area of the square, in cm^2?
   - A. 78
   - B. $78 - 18\sqrt{3}$
   - C. $78 + 18\sqrt{3}$
   - D. 84
   - E. $84 - 18\sqrt{3}$
   - F. $84 + 18\sqrt{3}$

Ans: E

22. A square piece of metal has a semicircular piece cut out of it as shown below. The area of the remaining metal is 100 cm^2.
Which one of the following is a correct expression for the length of the side of the square in centimetres?



   - A. $20\frac{\sqrt{2}}{\sqrt{8-\pi}}$
   - B. $10\frac{\sqrt{2}}{\sqrt{4-\pi}}$
   - C. $20\frac{\sqrt{2}}{\sqrt{8+\pi}}$
   - D. $10\frac{\sqrt{1}}{\sqrt{8-\pi}}$
   - E. $20\frac{\sqrt{1}}{\sqrt{4-\pi}}$

Ans: A

23. Solve the inequality.

$$x^2 \geq 8 - 2x$$

A. $x \geq 4$

B. $x \leq 2$ and $x \geq -4$

C. $x \geq -2$ and $x \leq 4$

D. $x \geq 2$ or $x \leq -4$

Ans: D

24. The total surface area of a cylinder, measured in square centimetres, is numerically the same as its volume, measured in cubic centimetres.

The radius of the cylinder is r cm, and the height is h cm.

Express h in terms of r.

A. $h = \frac{2r}{r-2}$

B. $h = \frac{2r}{r+2}$

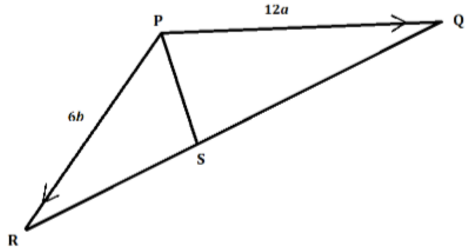
C. $h = r + 2$

D. $h = r - 2$

E. $h = 2r(r - 2)$

Ans: A

25. PQR is a triangle. $PQ = 12a$, $PR = 6b$.
S is a point on RQ, and the ratio $RS : SQ = 1 : 2$.
What is $\overrightarrow{PS}$ in terms of a and b?



A. $3b - 6a$
B. $3b + 6a$
C. $4b - 4a$
D. $4b + 4a$
E. $8b - 4a$
F. $8b + 6a$
G. $9b - 6a$

Ans: D

26. P is a regular 12-sided polygon.
The length of a straight line joining a vertex to the centre of P is 6 cm.
Q is a regular hexagon with side length x cm. The areas of P and Q are equal.
What is the value of x^2?

A. 12
B. 24
C. 36
D. 72
E. $12\sqrt{3}$
F. $24\sqrt{3}$
G. $36\sqrt{3}$
H. $72\sqrt{3}$

Ans: F

27. s is directly proportional to the square root of t.

    When $t = 0.04$, $s = 36$. What is t when $s = 45$?

    A. 0.0256
    B. 0.05
    C. 0.0625
    D. 0.09
    E. 0.16
    F. 0.25
    G. 0.3
    H. 0.5

    Ans: C

28. If you look at a 12-hour analogue clock and the time is $1:40$, what is the obtuse angle between the hour and the minute hands?

    A. 130°
    B. 150°
    C. 170°
    D. 190°
    E. 210°

    Ans: C

29. The original price of a coconut is x. The price is increased by 125%. The increased price is then decreased by 40% twice to y. Find the relationship between x and y.

   A. $\frac{3}{2}$
   B. $\frac{9}{20}$
   C. $\frac{27}{10}$
   D. $\frac{9}{25}$
   E. $\frac{9}{5}$
   F. $\frac{1}{5}$
   G. 1
   H. $\frac{81}{100}$

Ans: H

30. The square $ABCD$ is positioned so that its vertices are at the points with coordinates: $(1, 1), (-1, 1), (-1, -1), (1, -1)$.

   The square is rotated clockwise through 90° about the origin and then reflected in the line $y = x$.

   Which transformation will return the square to its original orientation?

   A. A reflection in the x-axis.
   B. A reflection in the y-axis.
   C. A reflection in the line $y = -x$.
   D. A rotation of 90° clockwise about the origin.
   E. A rotation of 90° anticlockwise about the origin.

Ans: B

Critical Thinking

1. Recorded crime figures – the figures which police authorities produce – have always been a poor way to identify crime trends. They are really a measure of police activity and priorities. A big operation to tackle knife crime, for instance, may uncover and record many more offences involving knives: it does not mean knife crime is rising. Added to that, there is an inbuilt temptation for police officers to 'adjust' their crime figures when targets need to be met. As with all recorded activity or performance data, there is always a risk of inaccuracy, confusion and fraud. A much more reliable measure of crime is the Crime Survey of England and Wales which produces figures by asking people if they have been victims of crime.

   Which one of the following is a conclusion that can be drawn from the above passage?

   A. The police regularly prioritise the tackling of crimes which help them meet their targets.

   B. The police care more about meeting targets than they do about tackling crime.

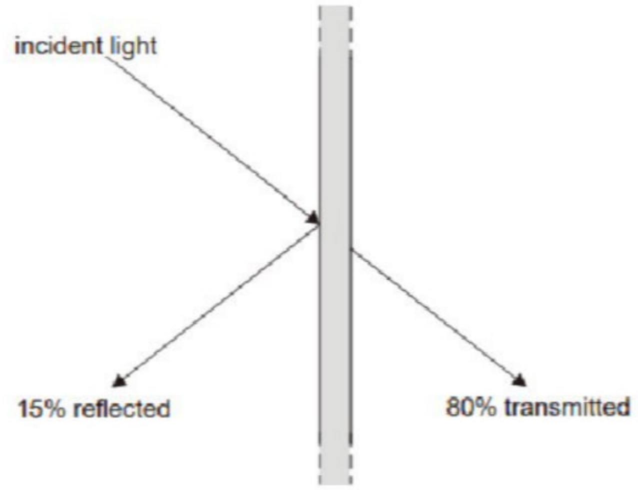
   C. Much less crime takes place than police figures tend to indicate.

   D. Victims of crime fear retaliation from criminals if they report them to the police.

   E. To find out whether crime levels are increasing we should survey people's experience of crime.

Ans: E

2. A certain type of window glass allows 80% of the incident light on it to pass through. 15 % is reflected back and 5% is absorbed.



If a double glazing panel is made from two sheets of this glass with a narrow gap in between, and used to glaze window, how much of the incident light from outside will pass into the room? (to the nearest 1%)

A. 64%

B. 65%

C. 79%

D. 80%

E. 95%

Ans: B

3. "The traditional school sports day – featuring races with clear winners and losers – is being replaced in some schools by "fun days" of non-competitive games. The rationale for this is that the self esteem of children who always lose in competitive races is damaged. But in adult life, competition for jobs, partners and social status is unavoidable. Schools should recognise this and revert to traditional sports days."

Which one of the following is an assumption of the above argument?

A. Children are not naturally competitive.

B. Children who lose competitive races will be unsuccessful in adulthood.

C. Non-competitive games boost children's self esteem.

D. School sports should prepare children for adult life.

Ans: D

4. "We worry that the pesticides used on crops may get into our food, but few people know that plants make natural pesticides to protect them against threats to their existence. Every day we eat fruit and vegetables containing these natural pesticides. Since our consumption of natural pesticides vastly outweighs that of synthetic pesticides, our health is at greater risk from natural pesticides than from synthetic ones."

Which one of the following, if true, most weakens the above argument?

A. Natural pesticides in some plants can harm animals.

B. Humans have evolved to tolerate natural pesticides in food crops.

C. Every natural pesticide is toxic if consumed in sufficient quantity.

D. Traces of synthetic pesticides have been found in fruit and vegetables.

Ans: B

5. One of the games at a charity fund raising event was 'Guess How Many Jelly Beans are in the Jar'. Prizes were awarded according to how close the guesses were to the exact number. The results of the game are shown below in the table.

Position	Guess	Winner
1st	125	Jessie
2nd	140	Saul
3rd	142	Imran
4th	121	Marie
5th	120	Hank

How many jelly beans were in the jar?

A. 129
B. 130
C. 131
D. 132
E. 133

Ans: D

6. Perfect pitch – the ability to identify any note of music without inferring it from a reference note – is usually found to be a characteristic only of people who were taught music before the age of 6. So teaching music to children under the age of 6 should become a priority in primary schools. This could mean that in the future the majority of the population would have perfect pitch.

Which one of the following describes a flaw in the above argument?

A. It assumes that adults could acquire perfect pitch if they were taught music.
B. It ignores the possibility that other factors may be necessary for acquiring perfect pitch.
C. It assumes that having perfect pitch is necessary for success as a musician.
D. It ignores the possibility that children under the age of 6 may not enjoy learning music.

Ans: B

7. The height and weight of three children were measured and the table below used to determine their BMI. Unfortunately some of the data have been lost. We know that Alex is 162cm tall, Jay is 150cm tall and has a BMI of 22, and Charlie is 156cm tall and has a BMI of 24. The combined weight of the children is 172kg.

Children's Metric BMI Tables

BMI

Height (cm)	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
144	26	29	31	33	35	37	39	41	43	45	47	49	51	53	55	58	60	62	64	66	68	70	72
147	28	30	32	34	36	38	41	43	45	47	49	51	54	56	58	60	62	64	66	69	71	73	75
150	29	31	33	35	38	40	42	45	47	49	51	54	56	58	60	63	65	67	69	72	74	76	78
153	30	32	35	37	39	42	44	46	49	51	53	56	58	60	63	65	67	70	72	74	77	79	81
156	31	34	36	38	41	43	46	48	51	53	55	58	60	63	65	68	70	73	75	77	80	82	85
159	32	35	37	40	42	45	48	50	53	55	58	60	63	65	68	70	73	75	78	80	83	85	88
162	34	36	39	41	44	47	49	52	55	57	60	62	65	68	70	73	76	78	81	83	86	89	91
165	35	38	40	43	46	49	51	54	57	59	62	65	68	70	73	76	78	81	84	87	89	92	95

Weight (kg)

What is Alex's BMI?

A. 25

B. 26

C. 27

D. 28

E. 29

Ans: A

8. Five children have had their height measured as part of a pre-school health check. Stuart is taller than Ruth who is taller than Margaret. Tim is shorter than Ruth and taller than Adrian.

Who must Adrian be shorter than?

A. Ruth, Margaret and Stuart

B. Stuart and Margaret, but not necessarily Ruth

C. Margaret, but not necessarily Stuart or Ruth

D. Ruth and Stuart, but not necessarily Margaret

E. Ruth, but not necessarily Stuart or Margaret

Ans: D

9. Competition between restaurants is fierce and restaurateurs are trying to gain recognition for the quality of their food. Research from Oxford University might just give them that edge over their competitors. More than a hundred people participated in a series of experiments to see if the taste of food was affected by the cutlery used to eat it. The study found that desserts tasted better when eaten using small spoons. Similarly yoghurt tasted creamier when eaten with a black spoon. Cheese cut with a heavy knife tasted more expensive than cheese cut with a lighter knife.

   Which one of the following is a conclusion that can be drawn from the above passage?

   A. People choose restaurants only because of the quality of the food.

   B. Restaurants should think carefully about what cutlery they use

   C. Restaurants should pay less attention to food presentation.

   D. Customers pay attention to the cutlery they are using.

   E. Variety of cutlery is more important than quality of food.

Ans: B

10. Luke walks his puppy to a nearby park and lets the puppy off the lead.

    As Luke starts walking along the path, the puppy runs on ahead, until it has gone 100 metres. The puppy then turns and runs back to Luke, who in that time has walked 50 metres. The puppy then goes on ahead again, turns after running 100 metres and runs back to Luke who has now walked another 50 metres.

    This routine continues until Luke has walked 1 km.

    If it takes 12 minutes for Luke to walk 1 km in the park, what is the puppy's average running speed during the same time?

    A. 12 km/h

    B. 5 km/h

    C. 10 km/h

    D. 14 km/h

    E. 15 km/h

Ans: E

11. The city council has proposed a plan to implement a congestion charge for vehicles entering the downtown area during peak hours. The proposal aims to reduce traffic congestion and improve air quality by encouraging commuters to use public transportation or carpool. It is also anticipated that the revenue generated from the congestion charge will be used to fund improvements in public transit infrastructure. Critics argue that the charge may disproportionately affect low-income individuals who rely on cars for their daily commute.

    What is the assumption underlying the city council's proposal to introduce a congestion charge for vehicles entering the downtown area during peak hours?

    A. Public transportation in the city is affordable and accessible for most commuters.
    B. Low-income drivers contribute the most to traffic congestion in downtown areas.
    C. Revenue generated by the congestion charge will be sufficient to improve public transit.
    D. Drivers will shift to alternative transportation methods to avoid the congestion charge.
    E. Air quality improvements will occur only if traffic congestion is completely eliminated.

Ans: D

12. A group of 5 men are able to load a pile of 90 logs onto a truck in 15 minutes.

    How long would it take 3 men to load 120 logs, if they load them at the same rate (answer to the nearest minute)?

    A. 12 minutes
    B. 18 minutes
    C. 33 minutes
    D. 35 minutes
    E. 42 minutes

Ans: C

13. The UK private security industry employs half a million people in a range of roles, including supervising public events, guarding cash in transport, and assisting police in surveillance work. Some government officials are pushing for the legal regulation of such work to be moved out of government agency control and placed in the hands of the private security companies themselves. According to these officials, self-regulation is justified on economic grounds and because of the sector's "willingness to ... be more accountable for its own actions." The foolishness of this argument is breathtaking. The power entrusted to these workers over valuable property—not to mention the safety of people—means that there's a fundamental need for monitoring and oversight of standards of conduct by an impartial, trustworthy source.

Which one of the following is an underlying assumption of the above argument?

A. The UK private security industry is likely to continue to grow rapidly in the near future.

B. Self-regulation of an industry is an unwise approach regardless of the sector concerned.

C. By government standards, the current regulatory system for the UK private security sector is economically efficient.

D. Many companies in the UK private security industry have a poor record when it comes to issues of professional misconduct or corruption.

E. The government agency staff are less susceptible to corrupt or unfair practices when regulating the workers than are those at the private security companies.

Ans: E

14. A three-dimensional shape is made by taking a cube and cutting off each of its corners. Assume that the cube is large and that the cuts are relatively small (extending to less than halfway along each edge).

How many edges does the new shape have?

A. 12

B. 18

C. 20

D. 32

E. 36

Ans: E

15. What is there in the universe that we know, or have good reason to believe, to possess consciousness? Humans, and perhaps certain other animals. What do they all have in common? They are living beings, made of flesh and blood. As far as we know, nothing else in the universe contains any consciousness at all. For all the mystery of consciousness, of exactly what it is, and of how and why it arises, there is one thing that is certain: it is a biological phenomenon. Any fears surrounding the possibility of so-called 'artificial intelligence' are therefore misplaced. If machines are not biological organisms, then they cannot have a mind. Do not expect man-made machines to be conscious—now or ever!

    Which one of the following best expresses the flaw in the above argument?

    A. It assumes that just because the only things we know to possess consciousness are biological organisms, they are the only things that could do.

    B. It draws a general conclusion about the properties of all naturally existing objects on the basis of merely a finite selection.

    C. It takes for granted that we understand exactly what it means to be conscious, when this is a vague and undefined term.

    D. It fails to consider the phenomenal speed with which new developments in technology are actually occurring.

    E. It ignores the possibility that some plants might possess a basic form of consciousness of their own.

    Ans: A

16. The current 'Special Offer' gimmick at Fareprice involves putting a band marked 'TWIN-PACK - 25% OFF' around pairs of selected items, such as washing powder.

    Which one of the following 'Special Offers' is exactly the same as Fareprice's?

    A. 'Buy one - Get a second for half price.'

    B. A larger packet containing 50% extra for no extra cost.

    C. 'Three for the price of two.'

    D. A coupon on each packet which gives 25% off the cost of the next packet.

    E. 'Five for the price of four.'

    Ans: A

17. There are countless ways to get rich in this world. Those who do not achieve wealth either do not find it important or simply have bad luck. Jane is poor but she has always valued material goods, so she must have been unlucky in her life.

    Which one of the following most closely parallels the reasoning used in the above argument?

    A. Anyone can become prosperous by working hard. Tom is not wealthy so he must not have been working hard.

    B. There are numerous ways to become famous. Among others, one can learn to play a musical instrument very well, study to become a world-class scientist, or just upload a funny video of oneself to the internet. Mike is not famous because he has not done any of these.

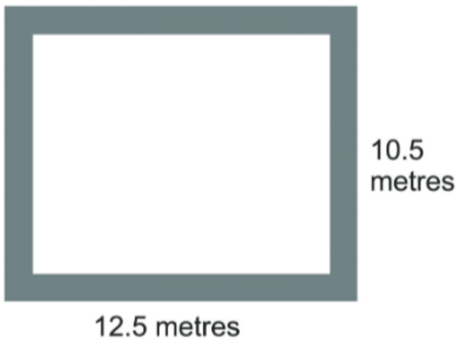
    C. To make a cake sweet, one can use either sugar or honey. This cake is not sweet because it was made with neither sugar nor honey.

    D. Two paths lead to the lake: one through the forest and another over the mountain. You must have arrived at the lake by the forest route because you were not on the mountain path.

    E. It takes a lot of luck to win the lottery. I have never won the lottery, so I must not have been lucky so far.

Ans: D

18. The whole of my garden is a patio, which is 12.5 by 10.5 metres. I intend to resurface my patio. I would like my patio to include a 25 cm border of bricks and the remainder of the patio inside the border to be covered with paving slabs.
    The bricks I will use for the border are 25 cm x 25 cm, and the paving slabs I will use for the remainder are 30 cm x 20 cm.
    What is the total number of bricks and paving slabs that I will need for my patio?



    A. 2000
    B. 2180
    C. 2184
    D. 2268
    E. 2272

Ans: B

19. Dark matter, a mysterious substance that does not emit, absorb, or reflect light, constitutes approximately 27% of the universe. Despite its invisibility, scientists infer its existence from its gravitational effects on galaxies and other celestial bodies. However, recent experiments attempting to detect dark matter particles directly have yielded inconclusive results. To advance our understanding, researchers must focus on developing more sensitive detection technologies and exploring alternative theories about the nature of dark matter.

What is the main conclusion of the passage?

A. Dark matter constitutes a significant portion of the universe and influences galaxies through gravity.

B. Scientists have not yet successfully detected dark matter particles directly.

C. Developing better detection technologies and exploring alternative theories is crucial to understanding dark matter.

D. Dark matter is a mysterious substance that cannot be observed directly through light.

E. Recent experiments on dark matter have been inconclusive.

Ans: C

20. A public telephone has been vandalised and will take only 50 pence coins and 10 pence coins. A woman puts £8.50 into the phone to make a long-distance call with twenty-five coins. Five 10 pence coins are returned when she finishes her call.

What combination of coins is retained by the machine?

A. Twice as many 10s as 50s.

B. One-and-a-half times as many 50s as 10s.

C. The same number of 50s and 10s.

D. Three times as many 50s as 10s.

E. One-and-a-half times as many 10s as 50s.

Ans: D

21. The ‘ticking time bomb’ scenario demonstrates that it is possible for torture to be morally permissible. In the scenario, a person with knowledge of an imminent terrorist attack that will kill many people is in the hands of the authorities and will disclose the information needed to prevent the attack only if she is tortured. The consequences of not torturing her—many people dying—are far worse than the consequences of torturing her—one person being tortured—so in this situation torture would be morally permissible.

Which one of the following is an underlying assumption of the above argument?

A. If an action is morally permissible in one situation, it is morally permissible in other situations too.

B. Dying is worse than torture.

C. The moral status of an action is determined by its consequences.

D. Ordinarily, torture would be morally wrong.

E. If an action is morally permissible, then it ought to be carried out.

Ans: C

22. Sometimes we are mistreated by others, and forgiveness is one kind of response to those who wrong us. However, sometimes we do things that appear to be morally wrong but, in fact, are not. If we are reproached, we can give an explanation for our action that justifies it. In these cases, we are claiming that, despite appearances to the contrary, what we did was morally permissible. Forgiveness and justification ought to be distinguished. When conduct is justified, the implication is that it was not morally wrong, but when conduct is forgiven, there is no such implication. What we are forgiven for is the morally wrong things we do.

Which one of the following best expresses the main conclusion of the above argument?

A. Sometimes we do things that appear to be morally wrong but, in fact, are not.

B. Forgiveness and justification ought to be distinguished.

C. What we are forgiven for is the morally wrong things we do.

D. Forgiveness is one kind of response to those who wrong us.

E. When conduct is justified, the implication is that it was not morally wrong.

Ans: B

23. Despite the government's promise to be more business-friendly, plans have been announced to change the law regarding the paid leave that couples are entitled to take following the birth of a child. This will allow new parents to take more time off work than under the current regulations. It is claimed that the current legislation makes it difficult for those with family commitments to manage all of their responsibilities, but it needs to be recognised that businesses (and in particular small businesses) need to operate in a way that guarantees that staff are available to do the jobs that need doing. The new proposals are undoubtedly going to make the system more complicated and could dissuade businesses from employing certain groups of people. They should be opposed.

    Which one of the following, if true, most weakens the above argument?

    A. The example given is only one of a range of measures being proposed.
    B. Many businesses now allow their employees to choose to work from home.
    C. The proposals from the government provide support for businesses to help with the cost of covering the work of staff taking leave.
    D. The problems associated with covering the work of absent members of staff become increasingly difficult the longer the absence goes on.
    E. The proposals will increase the amount of leave that employees are allowed to take in other circumstances as well as following the birth of a child.

Ans: C

24. A film star has agreed to sign autographs for £1 each at a charity function, all day from 9 am to 6 pm.

    He claims to sign his name consistently ten times every minute, but insists on having a ten-minute break after each full hour of signing.

    How much do the organisers hope that the autographs will raise?

    A. £4,500
    B. £4,600
    C. £4,700
    D. £4,800
    E. £5,400

Ans: C

25. Maths is the most popular subject taken at A Level in the UK. Most of the best-paid jobs are in the technology and finance sectors, which tend to require maths. The majority of students achieving the best grades in maths are boys, which helps to explain the persistence of the gender pay disparity. But what explains the gender disparity in Maths A Level results—genetic inheritance or social conditioning? International studies of achievement in maths show that in some countries girls outperform boys in maths, suggesting that social conditioning is the explanation. From this we can conclude that the better achievement in maths of boys in the UK is caused by low expectations of girls in maths by teachers and parents.

    Which one of the following best expresses the flaw in the above argument?

    A. It assumes that the only factors in social conditioning are the expectations of teachers and parents.

    B. It fails to explain why the relative achievement of boys and girls varies country by country.

    C. It fails to consider the changing picture of achievement by boys and girls over time.

    D. It ignores the relatively better achievement of girls in some other subjects.

    E. It confuses the gender pay disparity with achievement disparities.

Ans: A

26. Three friends are training to run a marathon. Part of this training involves running steadily round a track in the local park. They are all at different stages in their training schedules, so they run at different speeds. Alec completes each lap in 3 minutes, Barry takes 5 minutes, and Colin takes 6 minutes.

    They all leave the start line at the same time.

    If they each maintain their running speed throughout the session, how many more laps will Alec have completed compared to Barry when the three of them next cross the start line together?

    A. 2

    B. 4

    C. 5

    D. 6

    E. 10

Ans: B

27. Trains run regularly between Ayton and Exbay, stopping at Beetown, Chalton, and Deeford, according to the following timetable.

Departures			
Ayton	7:10	and every 40 minutes until	21:10
Beetown	7:25		21:25
Chalton	7:30		21:30
Deeford	7:44		21:44
Exbay	8:02		22:02

George lives a 14-minute walk away from Beetown station. He is going to the cinema in Exbay to watch a film that starts at 19:20. It will take him five minutes to walk from the station at Exbay to the cinema. George wants to make sure that he will miss no more than the first five minutes of the film.

What is the latest time George should leave home?

A. 17:36

B. 17:51

C. 18:05

D. 18:19

E. 18:31

Ans: B

28. One often hears that this or that action is justified or unjustified morally on the grounds that it is 'natural' or 'unnatural'. To say that something is 'natural' is really just to say that this is how things happen to be. Yet just because something happens to be the case, it doesn't mean that it ought to be the case. After all, there are plenty of things in life that are natural—for instance, illness and disease, droughts and earthquakes; even emotions such as aggression, jealousy, anger, and a desire for revenge. This does not mean that they are to be encouraged or welcomed. To say of something that it is natural, or unnatural, is not to provide anything by way of moral justification.

    Which one of the following illustrates the principle used in the above argument?

    A. The fact that human beings have a tendency to be competitive does not mean that this should be encouraged in schools.
    B. The fact that someone believes that human beings ought always to behave ethically does not mean that they actually will.
    C. The fact that some countries have laws protecting freedom of speech does not mean that all countries should.
    D. The fact that some things in life are clearly unpleasant does not mean that life itself is not worth living.
    E. The fact that someone thinks a particular person is not to be trusted does not mean that the person in question is indeed untrustworthy.

    Ans: A

29. In computer technology, a bit can take two values: 0 or 1. A byte is composed of 8 bits.

    I start with the following byte: 01000111.

    I choose two adjacent bits and change each of them to the opposite value (that is, a 0 becomes a 1, and a 1 becomes a 0). I repeat this procedure a number of times.

    Which one of the following bytes cannot be the result?

    A. 01001011
    B. 11111111
    C. 10101011
    D. 10111101
    E. 00001111

    Ans: C

30. In the main draw of a lottery, six of the balls (which are numbered from 1 to 49) are selected at random, then the numbers chosen are rearranged and displayed in ascending numerical order.

For instance: 3, 17, 20, 29, 34, 45.

In one draw recently, I noticed that the six numbers were made up of a total of ten digits, all different. The lowest number on this occasion was 1 and the highest number was 49, so the range of the six numbers was 48, the greatest possible range for the lottery main draw.

What is the smallest possible range when the six numbers in the main draw of the lottery have a total of ten digits, all different?

A. 21

B. 23

C. 27

D. 31

E. 32

Ans: D